



$$R_a = \frac{1}{L} \int_0^L |y_i| dx = \frac{1}{n} \sum_{i=1}^n |y_i|$$

$$R_z = \frac{1}{n} \left(\sum_{i=1}^n p_i - \sum_{i=1}^n v_i \right)$$

y_i	Local surface height
p_i	Local peak height
v_i	Local valley depth ($v_i < 0$)
R_a	Arithmetic mean roughness
R_z	Average roughness height
n	Number of sampling points
L	Sampling length

Fig. 2. Surface roughness profile (adapted from Cai et al. [26]), and definition of key statistical parameters [25].